University of Central Punjab
Faculty of Information Technology

PROGRAM (S) TO BE EVALUATED **BSCS**

A. Course Description

Course Code	CSNC 3413
Course Title	Information Security
Credit Hours	2+1
Prerequisites by Course(s) and Topics	Knowledge of information technology fundamentals (computer application, operating systems, applications, and networking) is required, or permission of the instructor is also required.
Assessment Instruments with Weights (homework, quizzes, midterms, final, programming assignments, lab work, etc.)	• Quizzes: 10% • Assignments: 15% • Class Participation: 10% • Mid Term: 25% • Final: 40%
Semester	6 & 7
Course Instructor (s)	
Course Instructor Email	
Course Coordinator	
Course Coordinator Email	Dr Ghulam Mustafa
Office Hours	Monday (09:00AM - 01:00PM), Tuesday (09:00AM - 01:00PM)
Office Location	C Block office no.3
Plagiarism Policy	• Cheating and plagiarism will not be tolerated in any course offered by UCP. Such activities/behavior will result in no credit for the assignment or examination and may lead to disciplinary actions. Please take the time to review the Rules & Regulations : https://ucp.edu.pk/rules-regulations/ • Such activities/behavior include copying (even with modifications) another student's work or letting your work be copied. Your participation in interactions with the instructor and your classmates is encouraged, but the work you submit must be your own. Collaboration is not permitted.
Tools Used in the Course	• Any General Purpose Programming Language (C/C++,Java, Python), PGP Plugin, Thunderbird, Zone Alarm or Windows Defender Firewall, SQL
Course Description	Students take this course to study the fundamentals of information security, both from a managerial and technological standpoint. Students gain an understanding of the various types of security incidents and assaults as well as techniques for preventing, detecting, and responding to such events. Additionally, students will study the fundamentals of using cryptography, a critical technology for implementing security features. Teams of students will present their research projects on subjects linked to information security at the final session.

Course Objectives	• To learn how to describe different information security threats and countermeasures. • To learn how to assess security issues and create defenses. • To learn how to describe the response to an information security event. • To be able to describe how Common Key and Public Key cryptography are used. • To be able to describe the mechanism used to safeguard the privacy and accuracy of data.			
Textbook (or Laboratory Manual for Laboratory Courses)	• Computer Security Fundamentals, 3rd edition by William Easttom • Cryptography and Network Security: Principles and Practice 7th Edition by William Stallings (Author) • Computer Security: Principles and Practice, 3/4th Edition by William Stallings (Author), Lawrie Brown (Author) • Security in computing, Charles P. Pfleeger and Shari Lawrence Pfleeger (2015), 5th edition, Prentice Hall, ISBN-13: 978-0132390774. • Understanding Cryptography, Christof Paar and Jan Pelzl (2009), 1st edition, Springer Publishing Company, ISBN: 3642041000 9783642041006.			
Reference Material	• Official (ISC)2 Guide to the CISSP CBK, 3rd edition • Principles of Information Security, Michael E. Whitman, and Herbert J. Mattord, Cengage Learning, ISBN: 1285448367 • Understand cryptography in-depth, Christof Paa and Jan Pelzl, Springer, ISBN: 3642041000 • Security Engineering, a guide to building dependable distributed systems, Ross J. Anderson (2020), 3rd Edition, Wiley Publications. • Building secure software, John Viega, and Gary McGraw (2011), How to avoid security problems the right way, Addison-Wesley Professional Computing Series, 1st edition, ISBN-13: 978-0321774958. • Applied Cryptography, Bruce Schneier, Protocols, Algorithms, and Source Code in C, 2nd Edition, Wiley, ISBN-13: 978-0471117094.			
Programming Assignments Done in the Course	N/A			
Class Time Spent on (in credit hours)	Theory			
	3ch			
Oral and Written Communications	Yes			

CLO	CLO STATEMENT	Bloom's Taxonomy Level	PLO
1	Identify and comprehend basics terms and technologies and concepts of information security	2	PLO 1-Academic Education
2	Investigate and analyze real situations from the information security point of view and model them using various security control measures	4	PLO 3- Problem Analysis
3	Apply the concepts of confidentiality, integrity, and availability into practice	3	PLO 4- Design/ Development of Solutions
1: Remember, 2: Understand, 3: Apply, 4: Analyse, 5: Evaluate, 6: Create			

Week	Lecture	Topics Covered	Evaluation Instrument	References	CLO
1	1	**Introduction of Information Security** ▪ Basics of Information Security ▪ CIA Model ▪ Basic terminologies		William Stallings Ch-1	1
	2	▪ Security attacks, mechanisms and services ▪ Models for network security		William Stallings Ch-1	1
2	3	**Classical Ciphers** ▪ Monoalphabetic Ciphers ▪ Caesar Cipher ▪ Monoalphabetic Substitution Cipher	Quiz #1	William Stallings Ch-2	1,2
	4	▪ Polyalphabetic Ciphers ▪ Vigenère Cipher ▪ Auto-key Vigenère Cipher		William Stallings Ch-2	1,2
3	5	▪ Vernam Cipher ▪ Onetime pad		William Stallings Ch-2	1,2
	6	▪ Frequency Analysis ▪ Playfair Cipher	Assignment #1	William Stallings Ch-2	2,3
4	7	• Hill Cipher • Extended Euclidean algorithm **Cryptanalysis** • Cryptanalysis of Caesar cipher	Quiz #2	William Stallings Ch-2	2,3
	8	**Classical Ciphers- Transpositions** • Rail Fence Cipher • Columnar Cipher		William Stallings Ch-2	1,2,3
5	9	**Stream Ciphers** ▪ RC4 Algorithm		William Stallings Ch-7	1,2,3
	10	**Block Ciphers** • Introduction of DES Algorithm • Claude Shannon Theory • Initial Permutation, Feistel Rounds • Expansion and reduction permutations		William Stallings Ch-6	1,2,3
6	11	▪ DES key generation ▪ Practical Question on DES		William Stallings Ch-6	1,2,3
	12	**Block Ciphers** ▪ Introduction to AES	Quiz #3		1

7	13	▪ AES Encryption	Assignment #2		1,2
	14	▪ AES Decryption and Key Scheduling			2,3
8	15	▪ Revision			
	16	Revision			
		Mid Term Week			
9	17	**Public Key Cryptography** ▪ Public and private key cryptosystems		Charles P. Pfleeger	3
	18	▪ RSA Algorithm ▪ RSA Encryption and Decryption		Charles P. Pfleeger	3
10	19	**Public Key Systems** ▪ Diffie Hellman Key Exchange	Quiz #4	Charles P. Pfleeger	1,2
	20	**Hash Functions** ▪ Hash Function Properties ▪ Applications of Hash Functions	Assignment #3	Charles P. Pfleeger	1,2
11	21	▪ Cryptographic hash functions ▪ Secure Hash Algorithm		Charles P. Pfleeger	1,2,3
	22	**Message Authentication Code** ▪ Hash based MAC		Charles P. Pfleeger	1,2,3
12	23	**Database Security** • GRANT and REVOKE		William Stallings	1,2,3
	24	**Operating System Security** ▪ Mandatory access control ▪ Discretionary access control	Quiz #5	William Stallings	1,2,3
13	25	**Network Security** ▪ Network Attacks ▪ Honeypots		William Stallings	1,2,3
	26	▪ Firewall ▪ Intrusion Detection Systems		William Stallings	1,2,3
14	27	▪ Firewall cont, ▪ Intrusion Detection Systems			1,2,3
	28	**Risk Management** ▪ Risk Identification and prioritization ▪ RE and RRL	Quiz #6		1,2,3

15	29	**Legal and Ethical Issues in Computer Security** ▪ Cyber Crimes and computer crimes ▪ Intellectual property			1,2,3
	30	▪ Protecting Program and Risk Assessment ▪ Ethical issues in computer security			
16		▪ Revision			
		▪ Revision			